

DSA0602 – Data Handling and Visualization

Unit II – Class Activity 1

Slot A

Comparative Evaluation of Statistical Visualizations Across Five Real-World Scenarios

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Introduction

This report addresses all five scenarios of Unit II – Class Activity 1 for DSA0602 (Data Handling and Visualization). For each scenario, a representative dataset was generated to reflect the situation described in the brief. Every required chart type was produced in R using ggplot2 (with dplyr for data preparation, treemapify for the treemap, and plotly for interactive dashboard versions), and the charts are compared against one another to identify which visualization most effectively communicates the insight the scenario asks for. The complete, runnable R source code accompanies this report as Assignment.R.

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1. Hospital Patient Admission Analysis

A synthetic dataset was built to represent one year of monthly patient admissions (August 2025 – July 2026) across five departments: Cardiology, Neurology, Pediatrics, Orthopedics and Emergency. Department-specific means and standard deviations were used so that Emergency carries the heaviest and most variable load, matching real hospital patterns, and two outlier surge months were injected (Emergency and Cardiology) to test how well each chart type surfaces anomalies.

Suggested libraries used: ggplot2, dplyr, treemapify, plotly (R)

1.1 Bar Chart – Total Admissions by Department

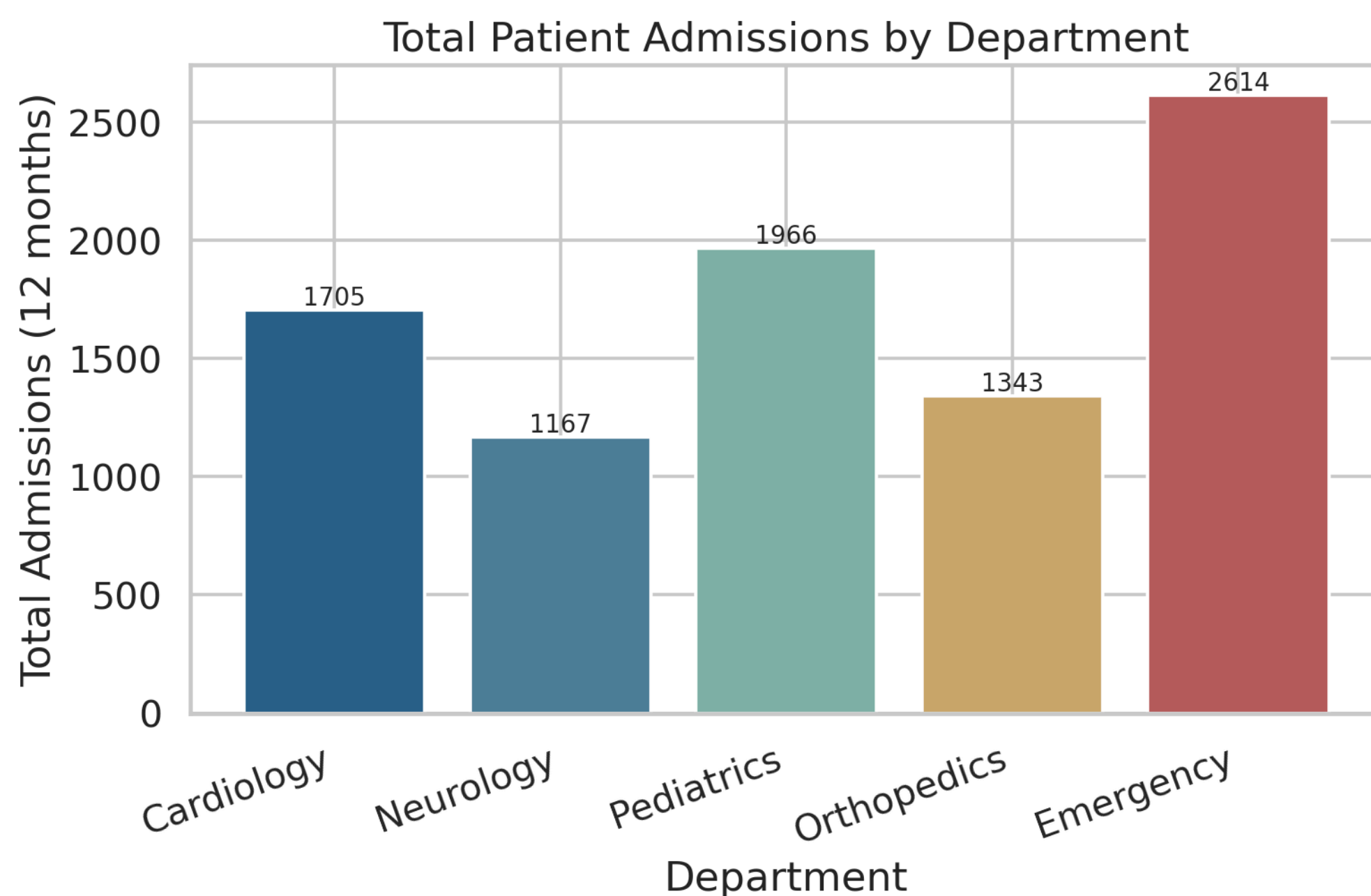


Figure 1.1: Total admissions per department, ranked highest to lowest.

1.2 Stacked Bar Chart – Quarterly Admissions by Department

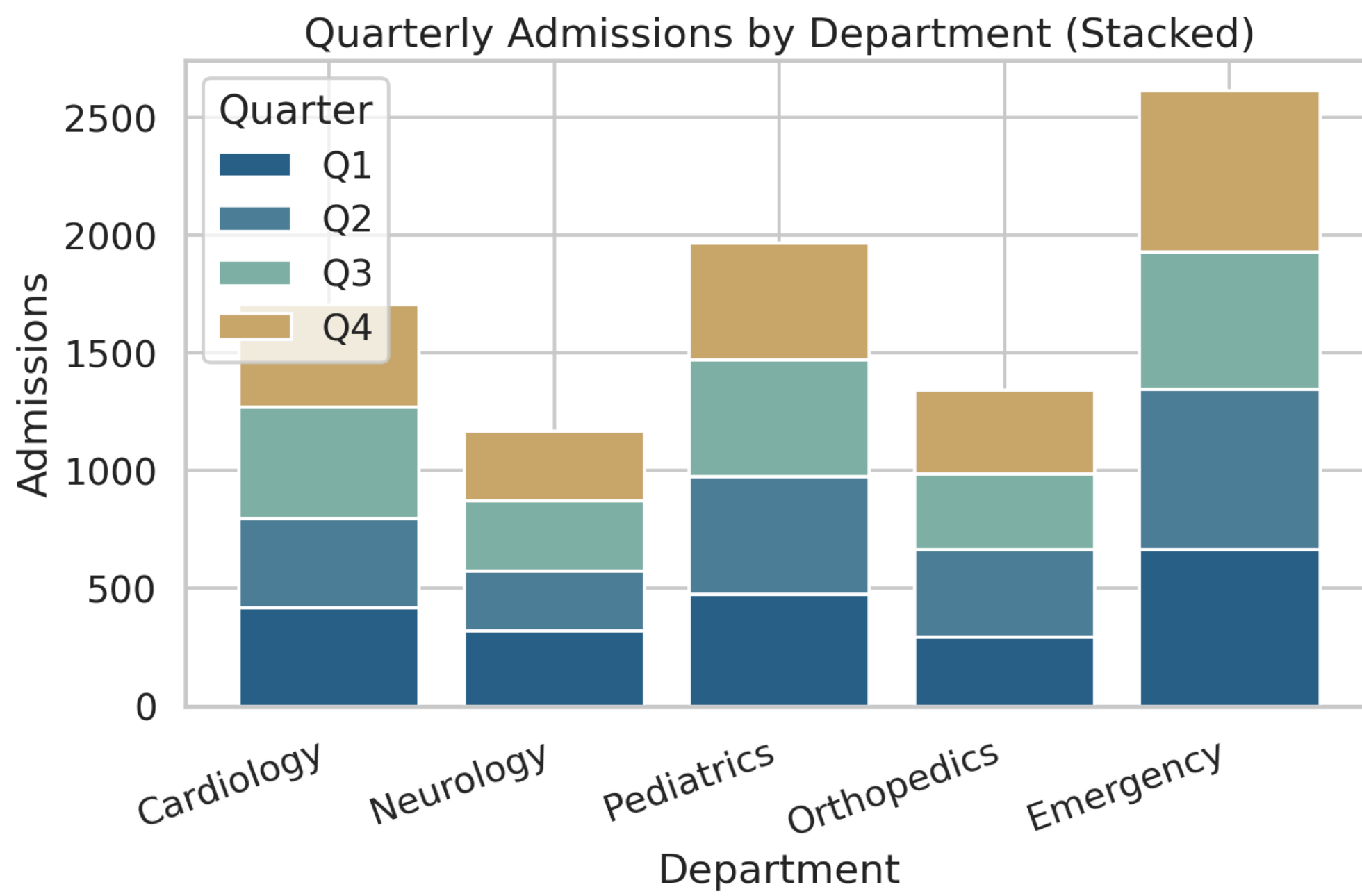


Figure 1.2: Each department's admissions broken down by quarter.

1.3 Treemap – Share of Total Admissions

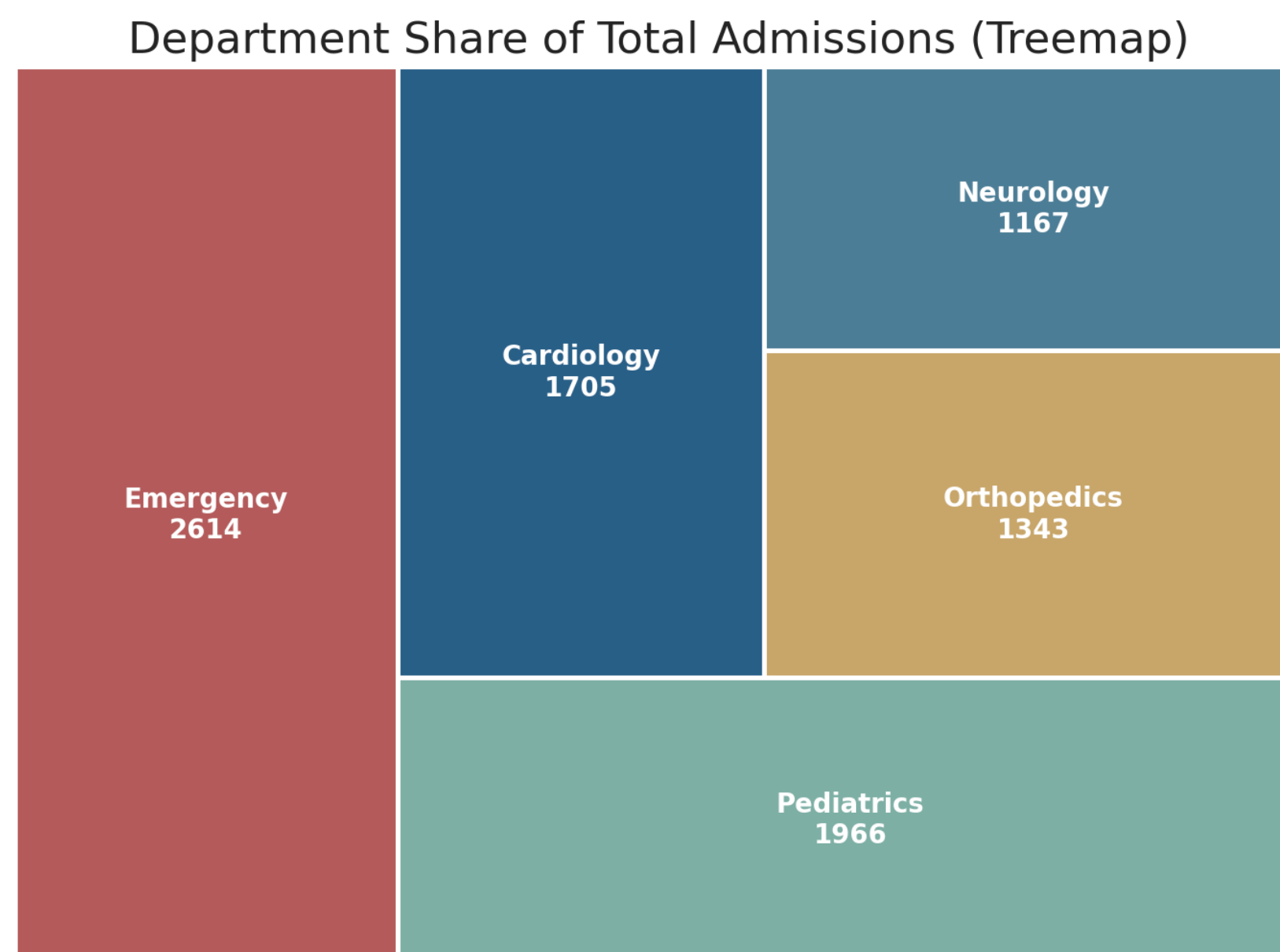


Figure 1.3: Proportional share of total patient volume by department.

1.4 Box Plot – Admission Variability by Department

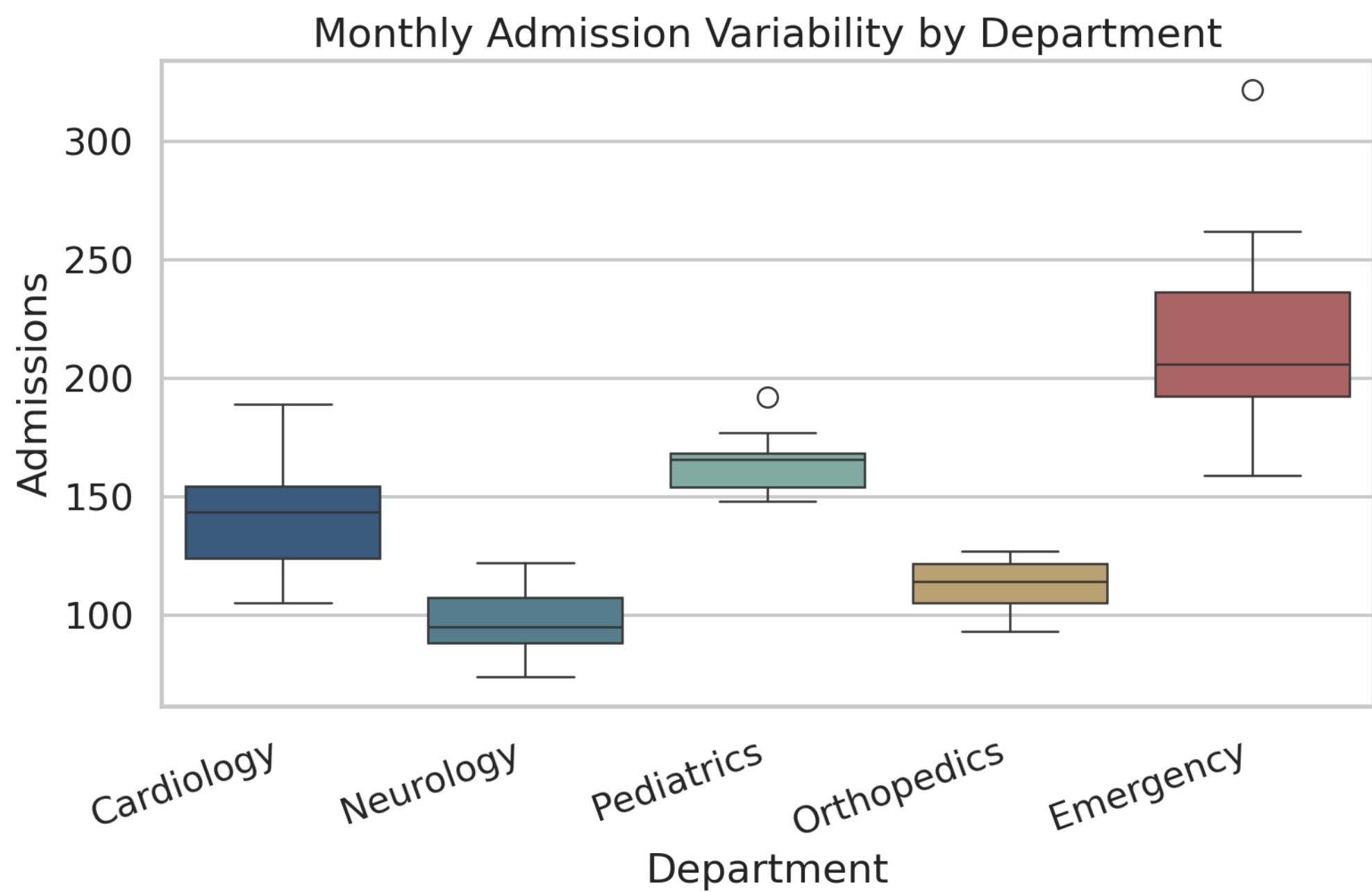


Figure 1.4: Median, interquartile range and outlier months per department.

1.5 Histogram – Distribution of Monthly Admission Counts

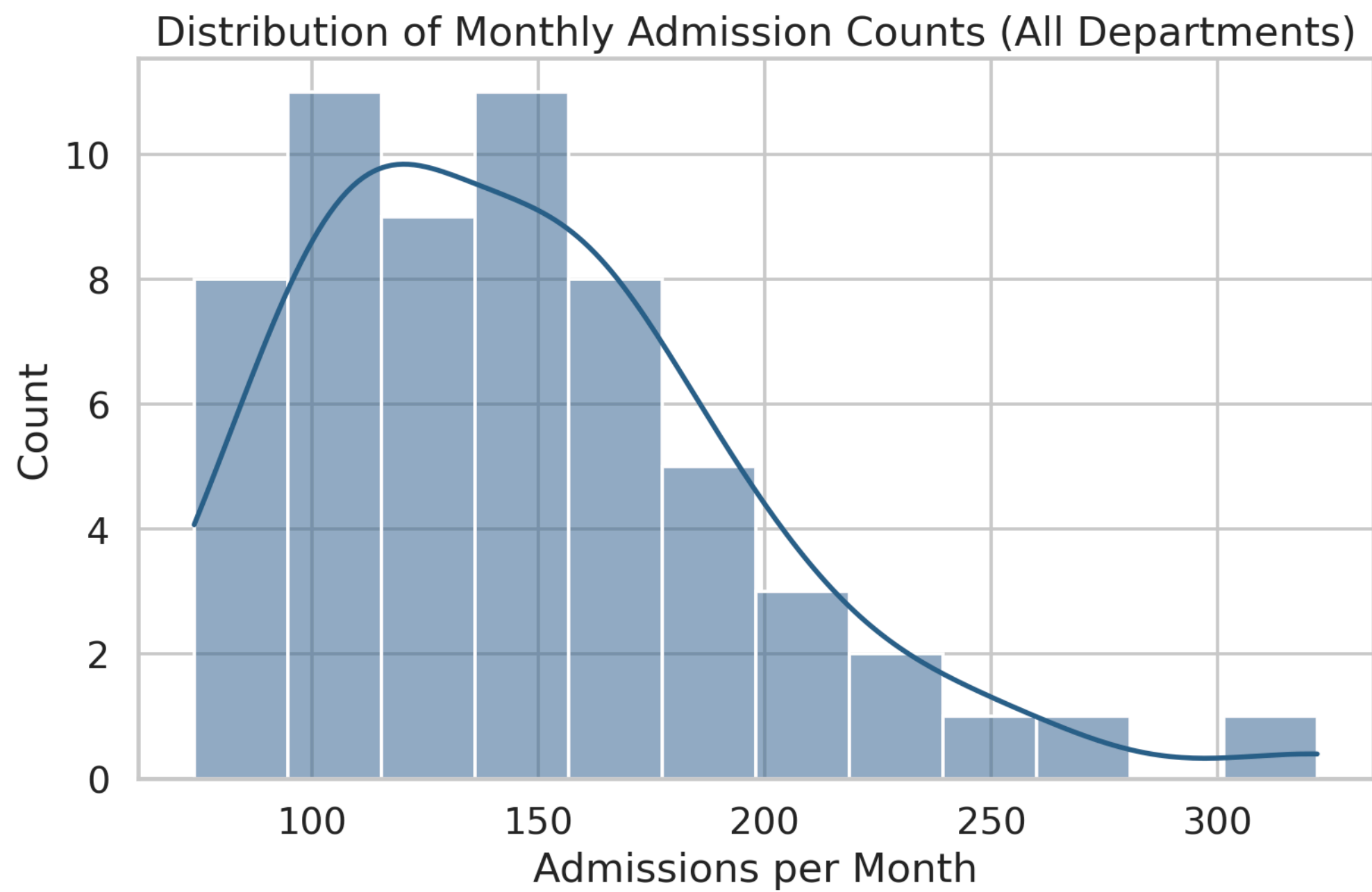


Figure 1.5: Overall shape of monthly admission counts across all departments.

1.6 Evaluation

Which visualization communicates workload, variability and distribution best?

- The bar chart (Fig. 1.1) is the clearest for ranking overall WORKLOAD – Emergency and Pediatrics carry the largest annual totals, visible in a single glance.
- The box plot (Fig. 1.4) is the strongest for ADMISSION VARIABILITY: it shows Emergency has both

the highest median and the widest interquartile range, plus a flagged outlier month, while Neurology is comparatively stable.

- The treemap (Fig. 1.3) is the most efficient for communicating each department's PROPORTIONAL SHARE of total patient volume to a non-technical audience (e.g. hospital board).
- The stacked bar chart (Fig. 1.2) is useful for tracking how each department's load is distributed across the year but becomes harder to read as more quarters/segments are added.
- The histogram (Fig. 1.5) shows the overall distribution is right-skewed, driven by Emergency's high counts, which is useful for capacity-planning statistics but does not separate departments.

Recommendation: Use the box plot as the primary chart for capacity planning, since variability and outlier months are what operations managers need to act on; pair it with the bar chart for a quick workload ranking and the treemap for executive-level summaries.

2. E-Commerce Customer Spending Behaviour

A synthetic dataset of 600 customers was generated with purchase amount (log-normally distributed, as is typical for spending data), order frequency, and product category (Electronics, Fashion, Home & Living, Beauty, Groceries). Category-specific multipliers were applied so that Electronics customers tend to spend more per order than Groceries customers, while Groceries/Beauty customers order more frequently.

Suggested libraries used: ggplot2, dplyr, scales, plotly (R)

2.1 Histogram – Spending Distribution

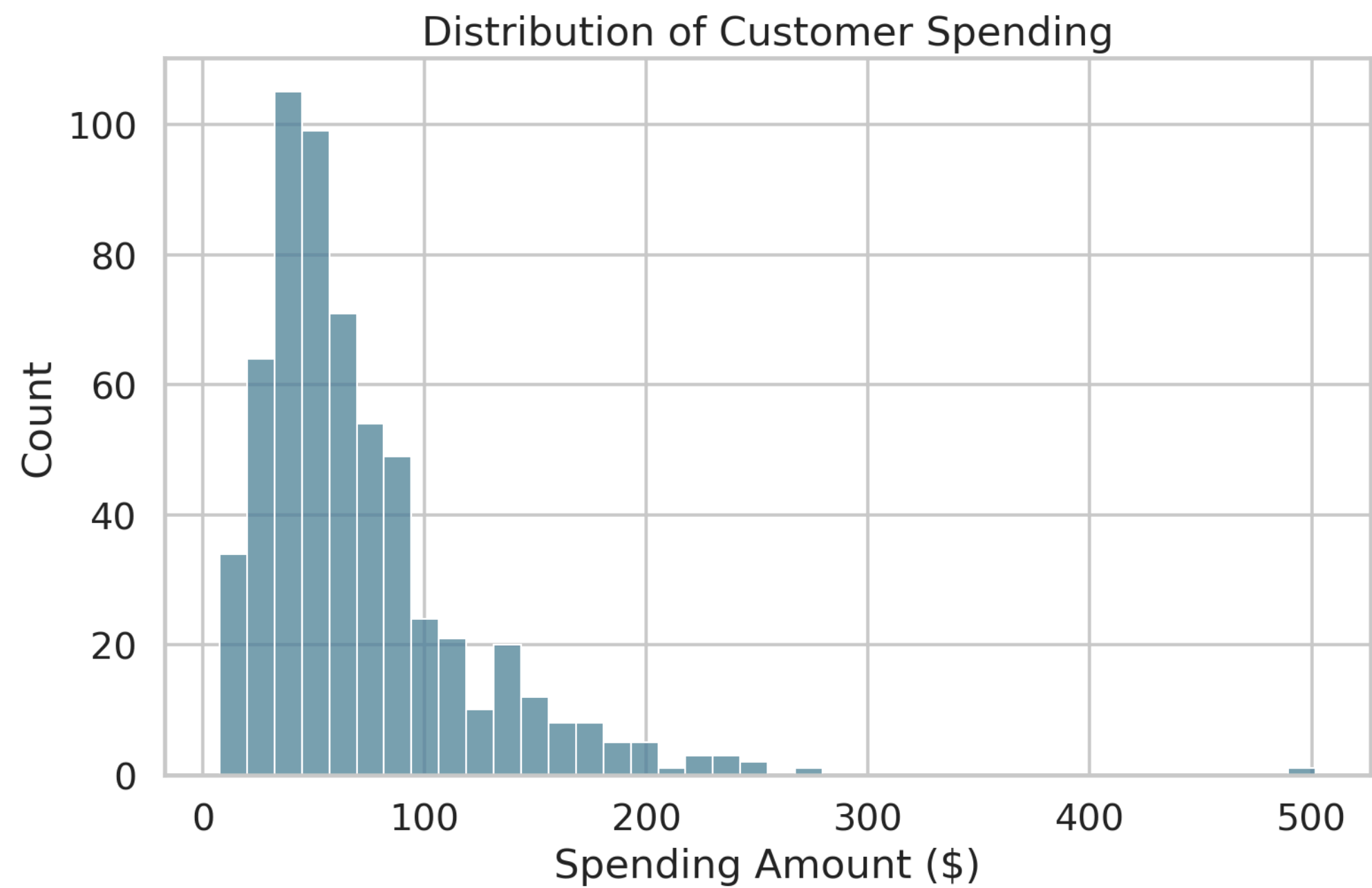


Figure 2.1: Distribution of individual order spending amounts.

2.2 Violin Plot – Spending by Category

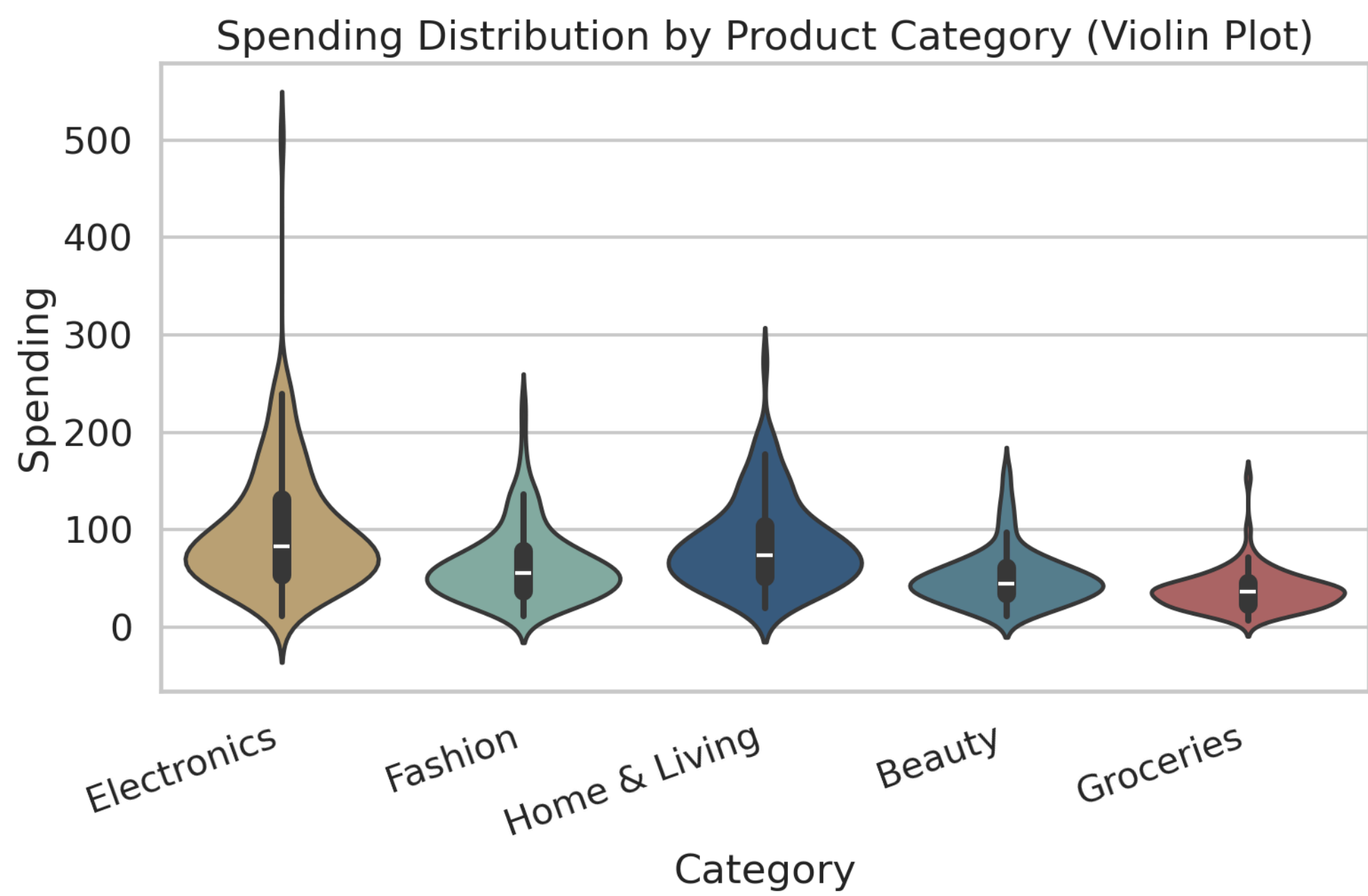


Figure 2.2: Full distribution shape of spending within each category.

2.3 Box Plot – Spending by Category

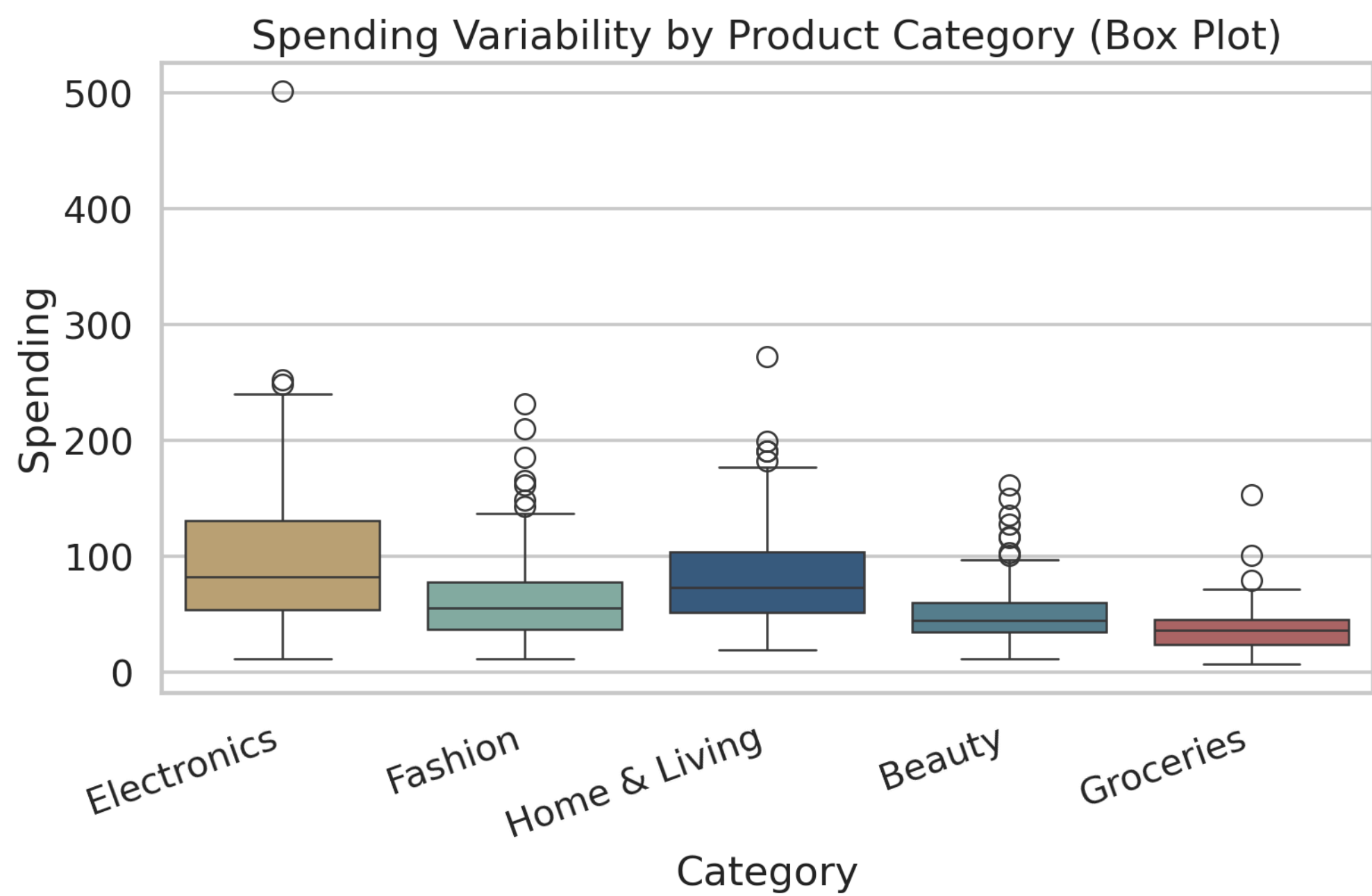


Figure 2.3: Median, spread and high-value outlier customers per category.

2.4 Density Plot – Overall Spending Distribution

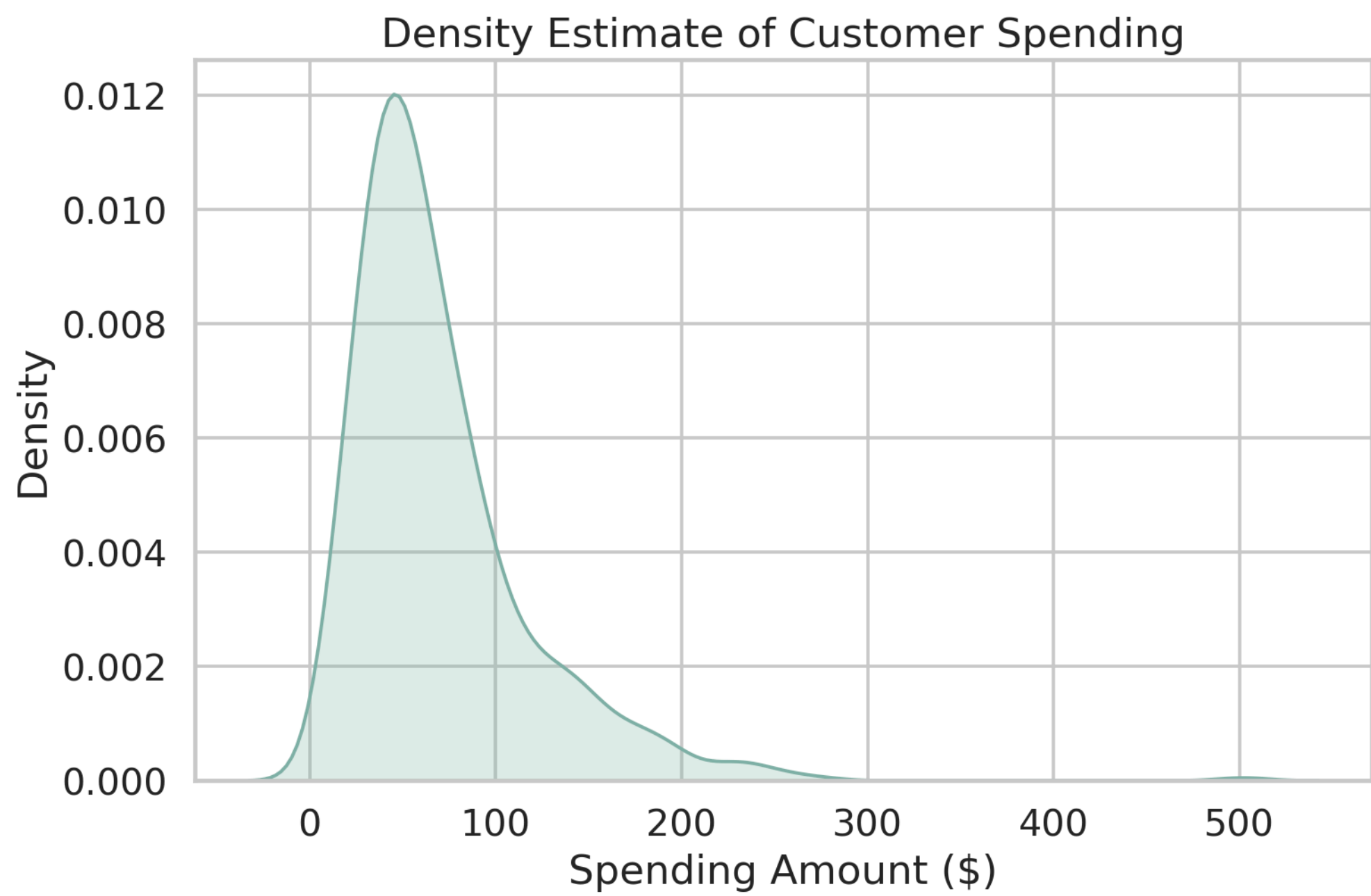


Figure 2.4: Smoothed density estimate of spending across all customers.

2.5 Bar Chart – Average Spending by Category

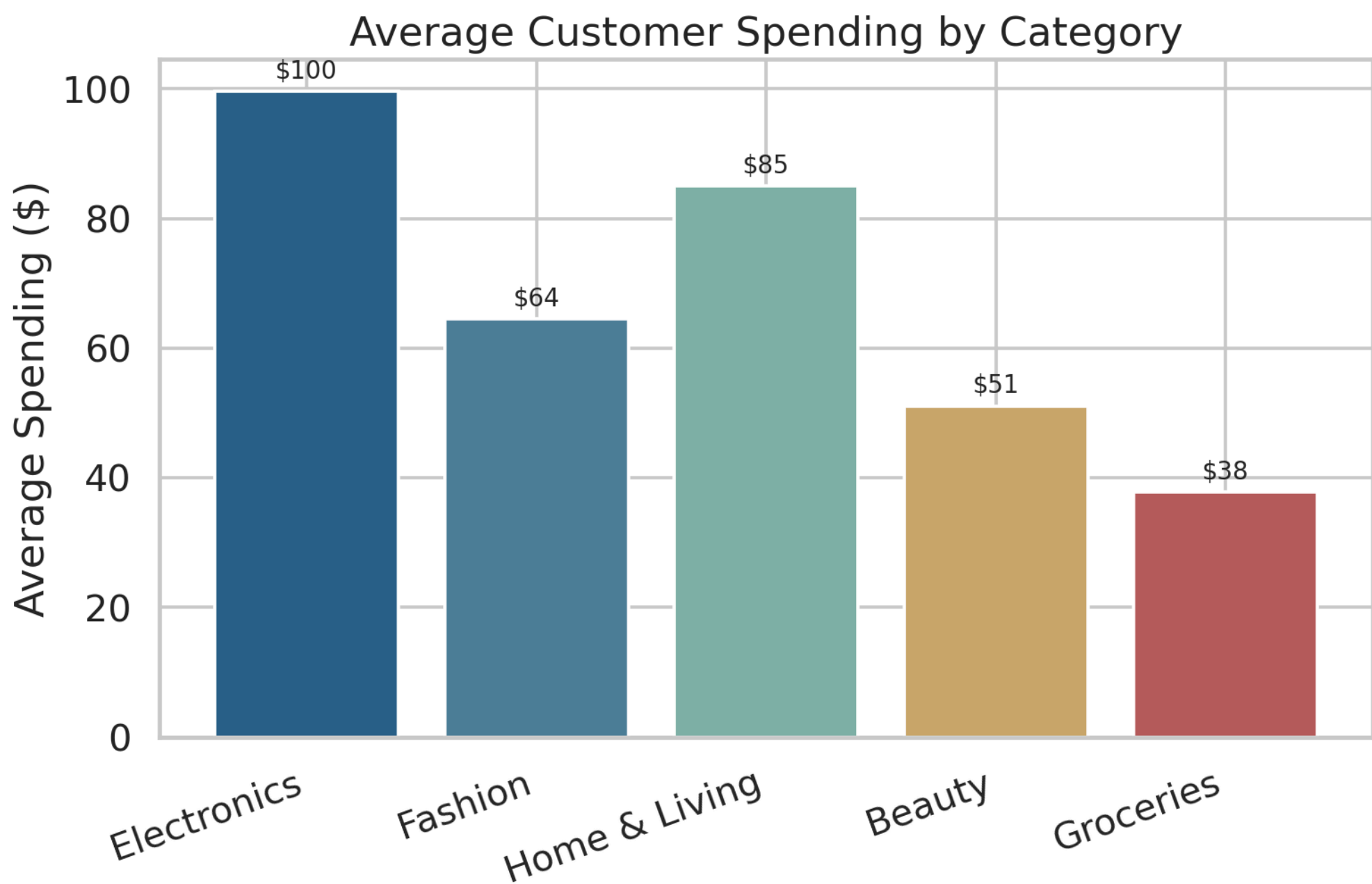


Figure 2.5: Mean spending per category, the simplest executive-facing summary.

2.6 Evaluation

Which visualization best identifies high-value customers, variability and trends?

- The box plot (Fig. 2.3) most clearly flags HIGH-VALUE CUSTOMERS: points above the upper whisker in Electronics and Home & Living are immediately visible as outliers worth targeting for loyalty offers.

- The violin plot (Fig. 2.2) best reveals SPENDING VARIABILITY and shape — e.g. Electronics shows a long right tail while Groceries is tightly concentrated at low spend.
- The histogram and density plot (Figs. 2.1, 2.4) confirm the overall spending distribution is right-skewed, which explains why the mean is pulled upward by a small number of big spenders.
- The bar chart of category averages (Fig. 2.5) is the fastest single-number PURCHASING TREND summary but hides the skew and the outlier customers entirely.

Recommendation: For business executives, the bar chart of average spending by category is the most appropriate — it answers 'where should we focus' in one glance. For the analytics team identifying which individual customers to target, the box plot is more appropriate because it exposes the outliers a simple average conceals.

3. University Examination Performance Dashboard

Marks for 80 students in each of five departments (Computer Science, Mechanical, Electrical, Civil, Business) were simulated using department-specific means and standard deviations, with a small number of very low scores injected into each department to represent at-risk students that a dashboard should surface.

Suggested libraries used: ggplot2, dplyr, plotly (R)

3.1 Box Plot – Marks by Department

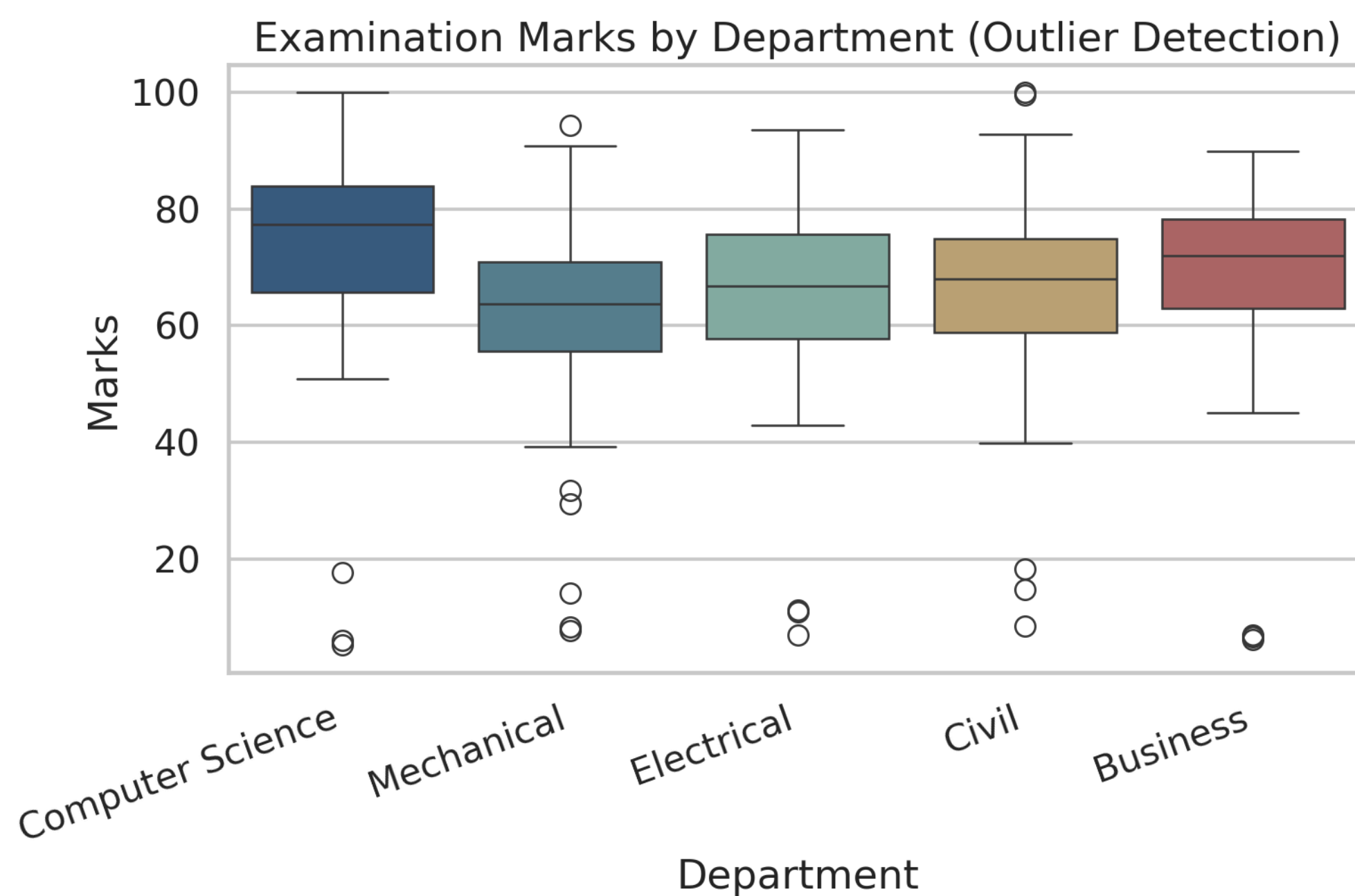


Figure 3.1: Median, spread and outlier (at-risk) students per department.

3.2 Histogram – Overall Marks Distribution

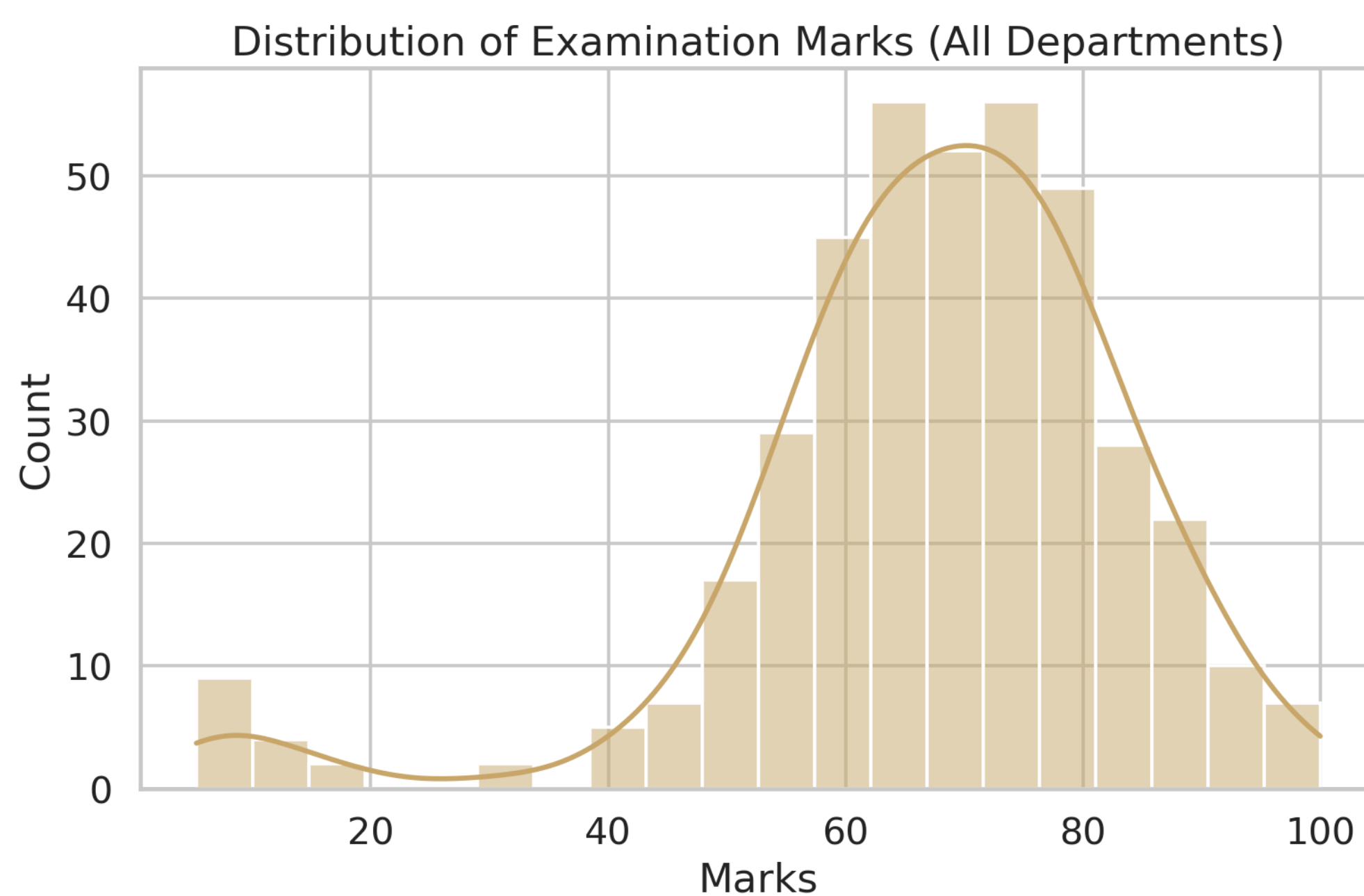


Figure 3.2: Distribution shape of marks across the full cohort.

3.3 Bar Chart – Average Marks by Department

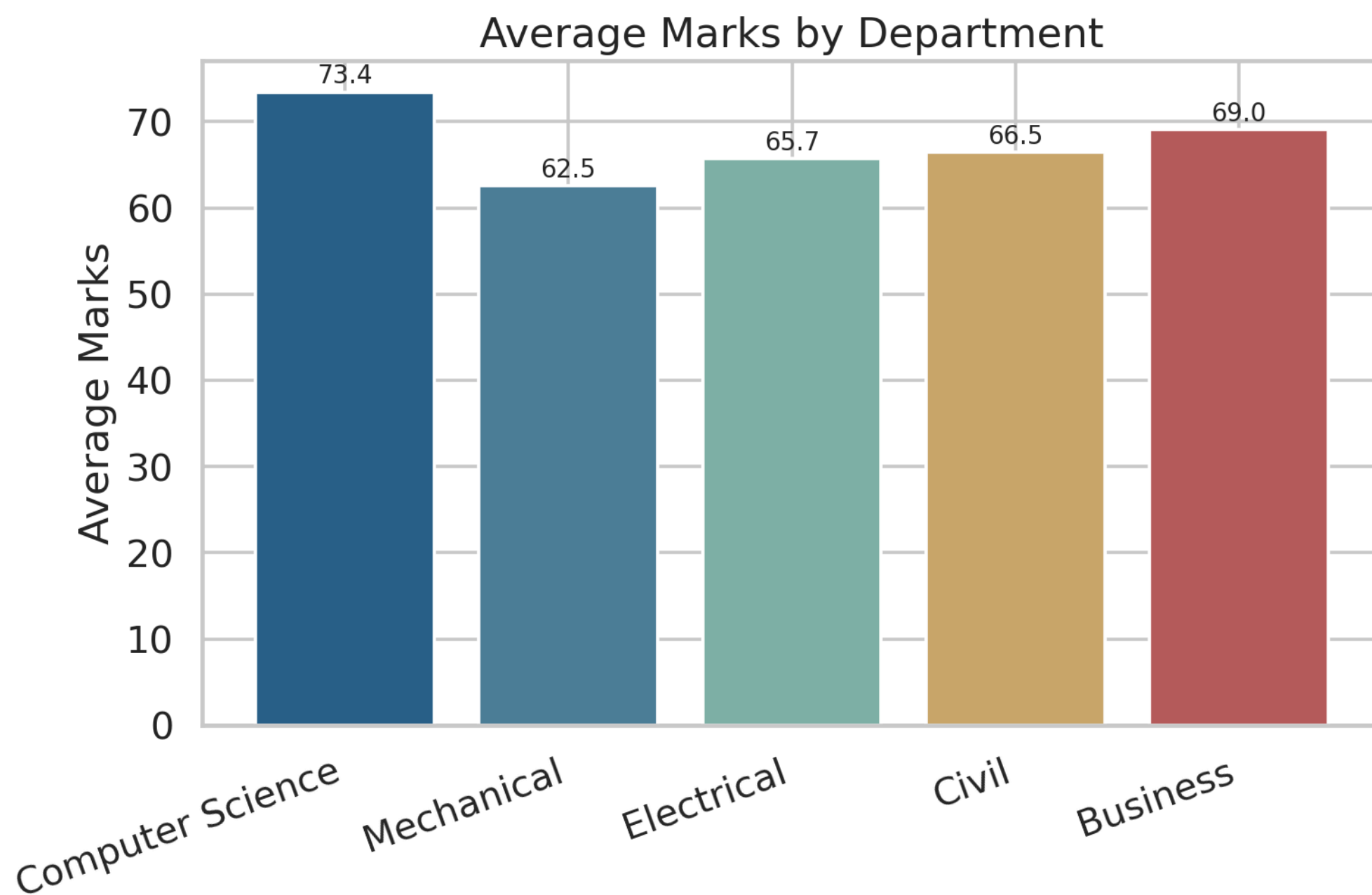


Figure 3.3: Departments ranked by mean marks.

3.4 Violin Plot – Shape of Mark Distributions

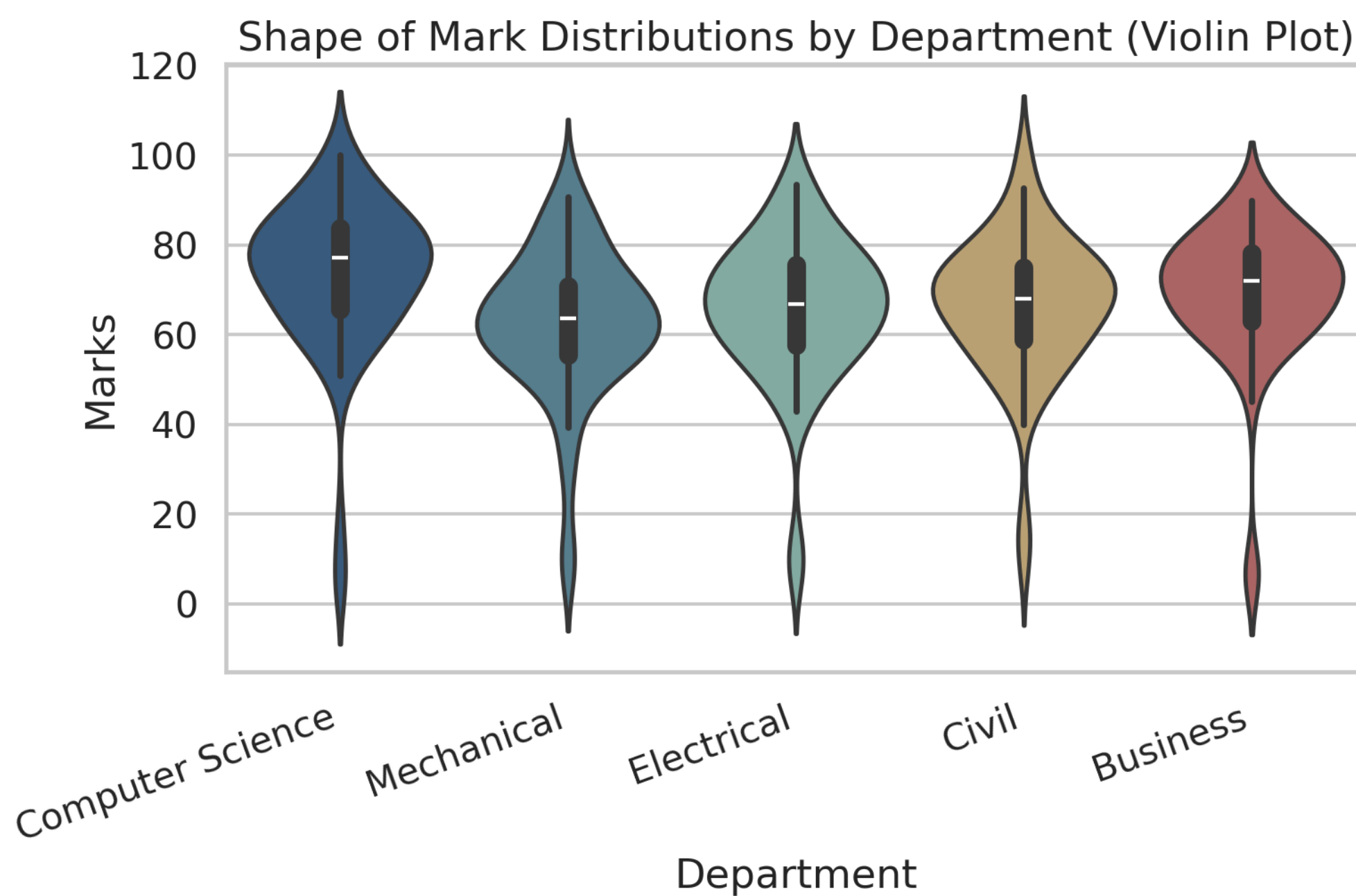


Figure 3.4: Full distribution shape of marks within each department.

3.5 Evaluation

Which charts give the clearest insight for academic decision-making?

- The box plot (Fig. 3.1) gives the CLEAREST overall insight: it shows Mechanical has the lowest median and widest spread, and identifies specific outlier (at-risk) students in every department in one compact view.
- The bar chart of averages (Fig. 3.3) is the quickest way to rank TOP-PERFORMING CLASSES (Computer Science highest, Mechanical lowest) but conceals variability and hides at-risk students entirely.
- The histogram (Fig. 3.2) is useful for checking whether the cohort's marks approximate a normal distribution, informing whether standard grading curves are appropriate.
- The violin plot (Fig. 3.4) adds distribution-shape detail beyond the box plot (e.g. bimodality) but is less familiar to non-technical academic staff and is harder to compare at a glance.

Recommendation: Recommend the box plot as the primary dashboard chart for department-level academic review, since it combines ranking, variability and at-risk-student identification in a single figure; retain the bar chart as a simple 'leaderboard' panel for quick reporting to leadership. Dashboard improvement: add a filter to toggle between semesters and a threshold line marking the pass mark directly on the box plot.

4. Climate and Rainfall Distribution Study

Monthly rainfall, temperature and humidity were simulated for four Indian cities (Chennai, Mumbai, Delhi, Bengaluru) following a monsoon-shaped seasonal curve, scaled per city to reflect realistic regional differences (e.g. Mumbai receiving substantially higher rainfall than Delhi).

Suggested libraries used: ggplot2, dplyr, plotly (R)

4.1 Histogram – Rainfall Distribution (All Cities)

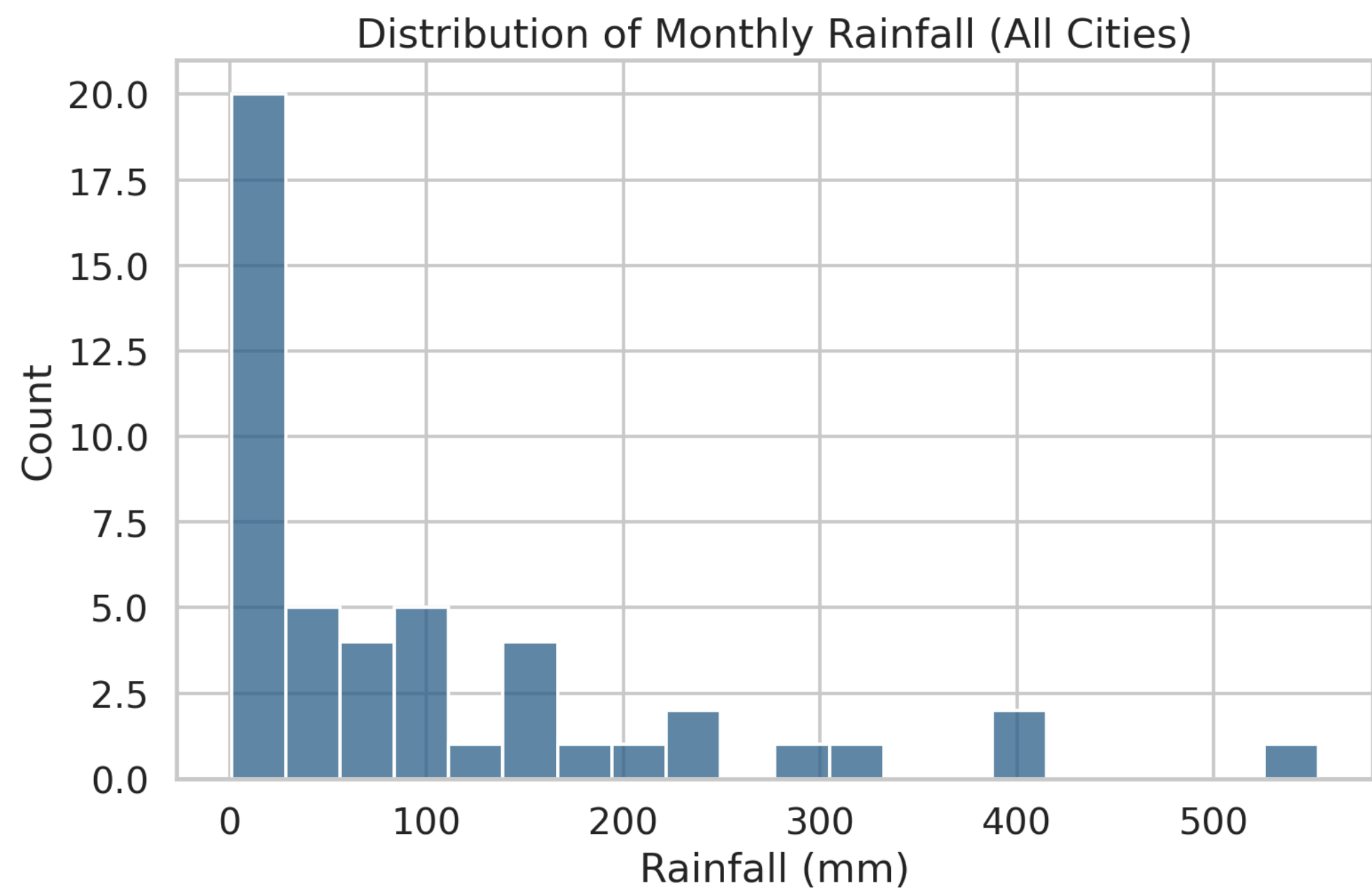


Figure 4.1: Distribution of monthly rainfall values pooled across all cities.

4.2 Density Plot – Rainfall Distribution by City

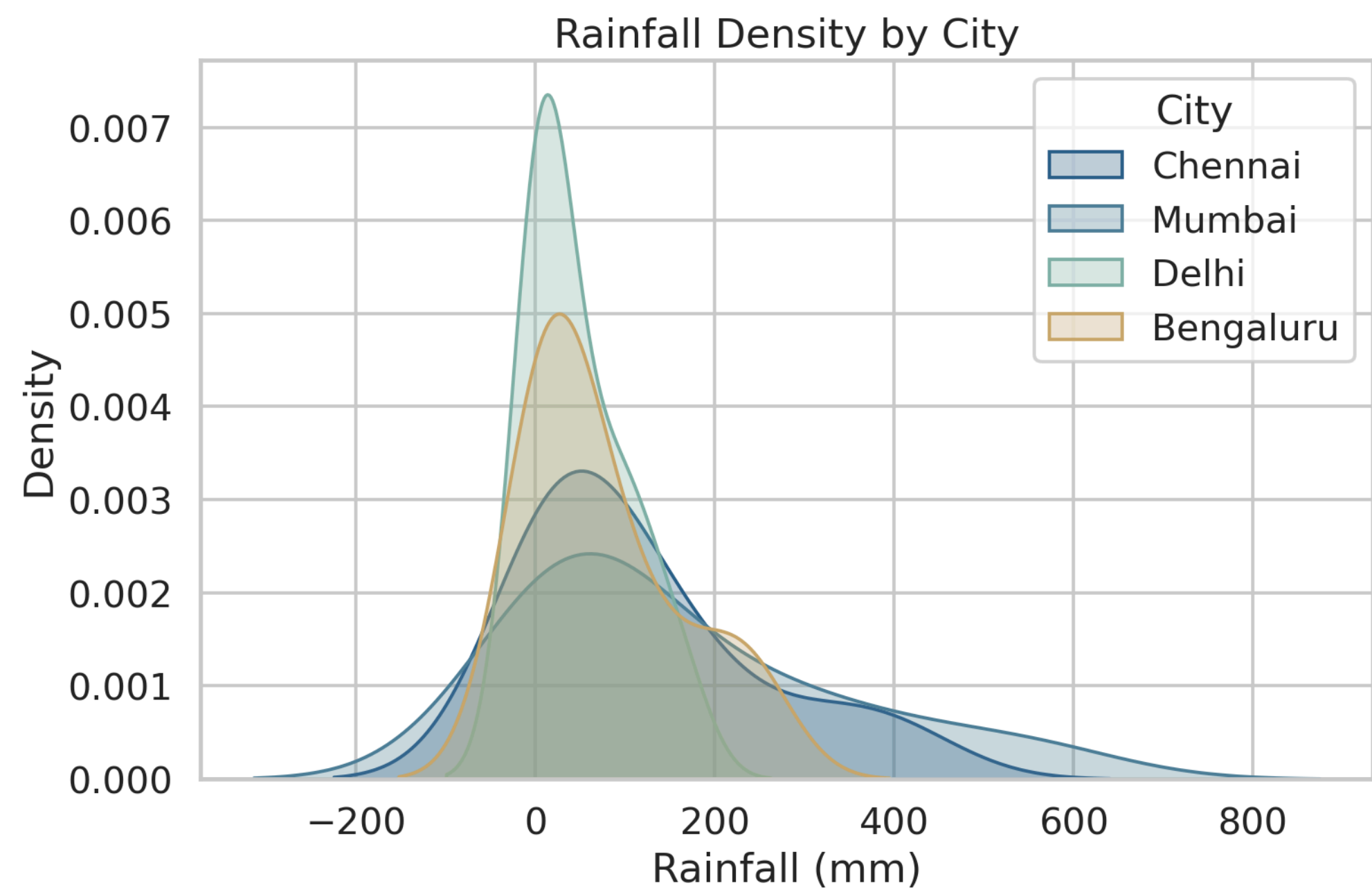


Figure 4.2: Smoothed rainfall distribution compared across cities.

4.3 Box Plot – Rainfall Variability by City

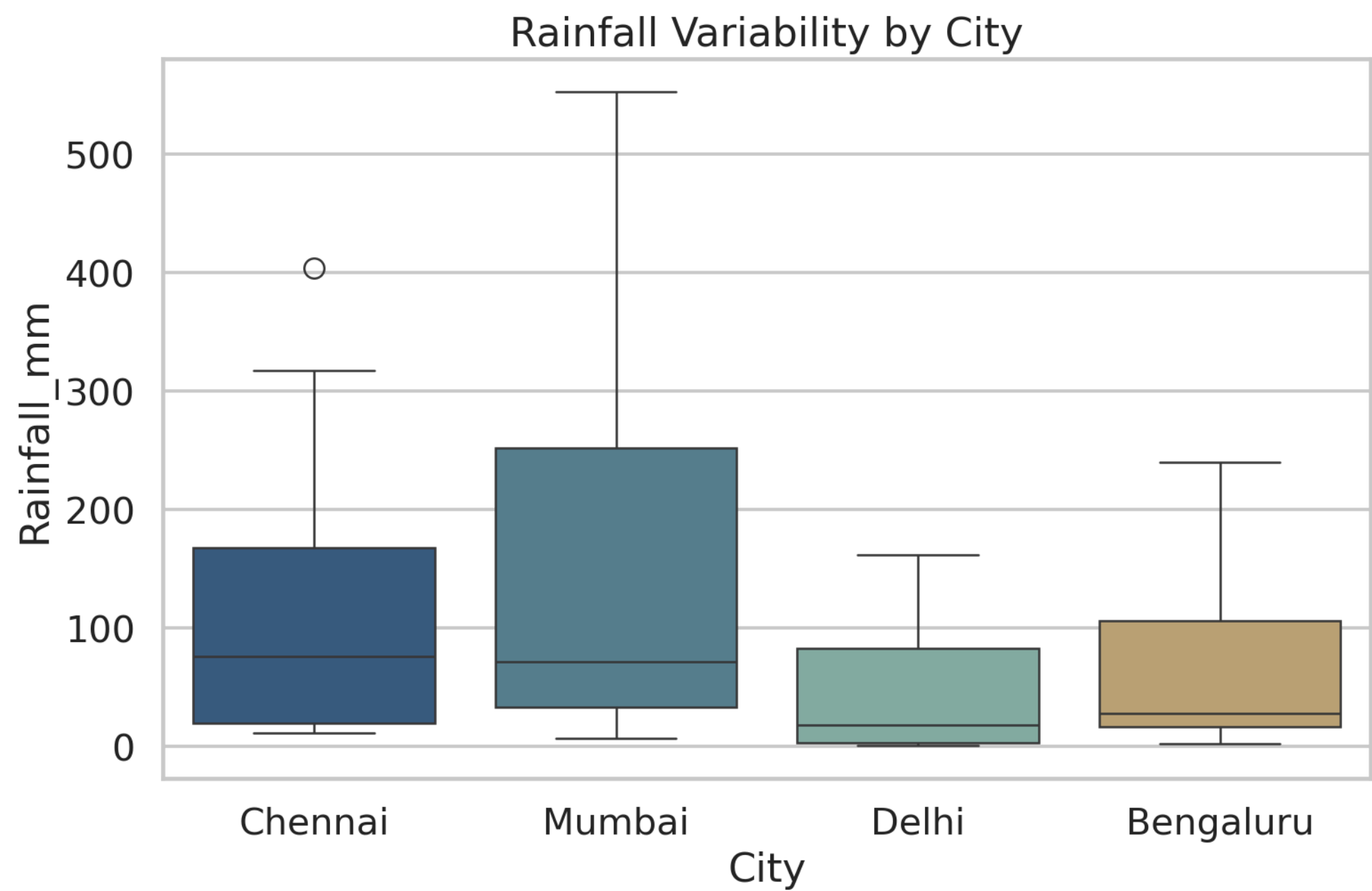


Figure 4.3: Median and spread of monthly rainfall per city.

4.4 Bar Chart – Seasonal Rainfall Pattern by City

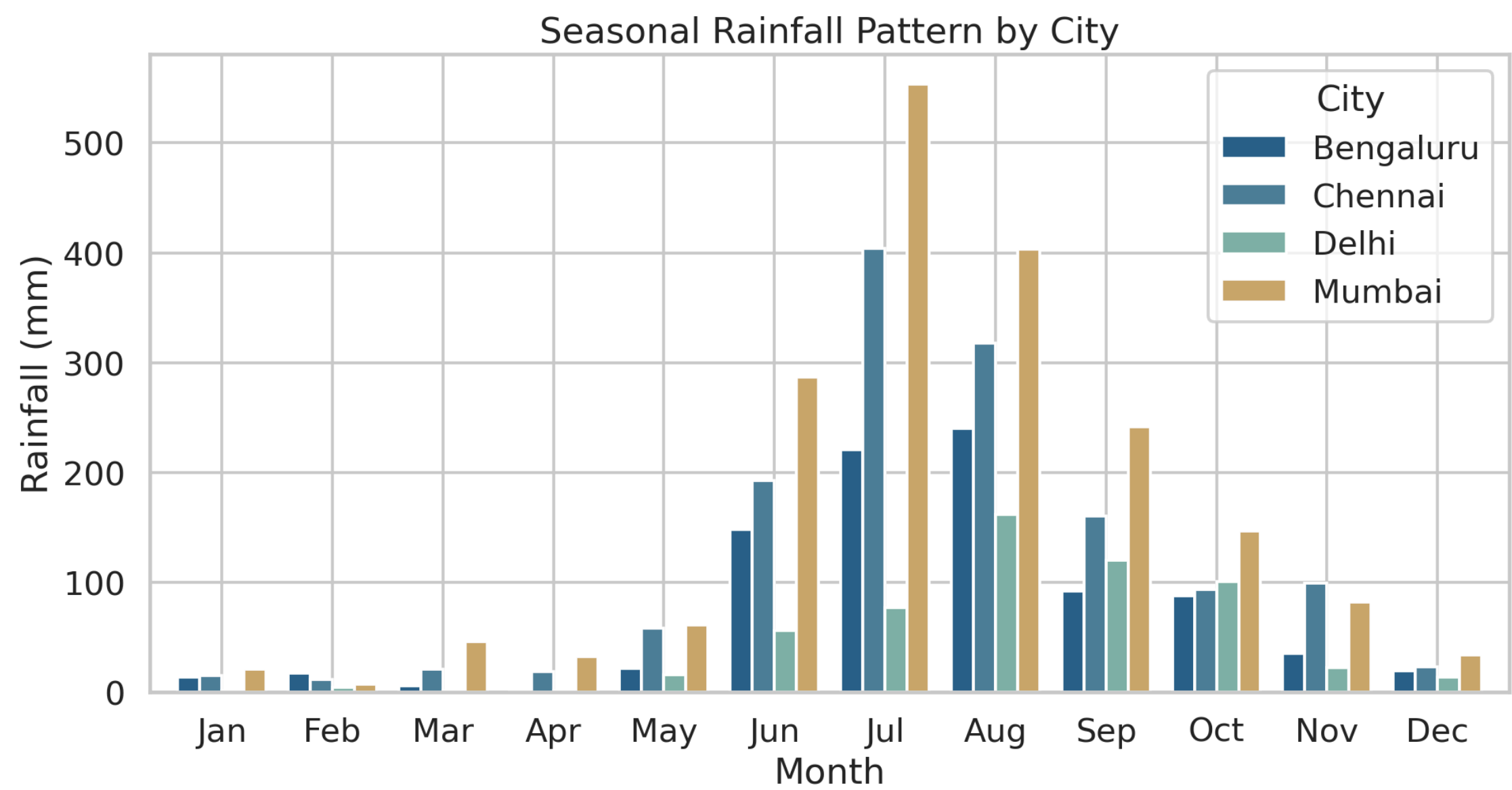


Figure 4.4: Month-by-month rainfall, grouped by city, preserving seasonal order.

4.5 Evaluation

Which visualization best represents seasonal variation and climate distribution?

- The bar chart grouped by month (Fig. 4.4) is the single best chart for SEASONAL RAINFALL

VARIATION because it is the only one that preserves the calendar/time axis, making the monsoon peak (Jun–Sep) and dry season immediately visible per city.

- The density plot (Fig. 4.2) best compares overall CLIMATE DISTRIBUTION shape between cities — Mumbai's curve is visibly right-shifted and wider than Delhi's, indicating both higher and more variable rainfall.
- The box plot (Fig. 4.3) gives a compact variability summary per city (median, IQR) but, like the histogram, discards the month-by-month/time ordering that matters most for a seasonal study.
- The histogram (Fig. 4.1), pooling all cities together, mainly shows that most months are low-rainfall with a long right tail from monsoon months — useful context but not city-specific.

Recommendation: For environmental researchers studying seasonality, the grouped bar chart (Fig. 4.4) is the most suitable primary visualization because it retains the temporal structure essential to a seasonal study; the density plot (Fig. 4.2) is the best secondary chart for comparing overall city-level climate distributions.

5. Bank Loan Risk Assessment

A synthetic portfolio of 700 loan customers was generated with income (log-normal, reflecting real income skew), credit score, loan amount, and repayment ratio. Customers were segmented into four risk bands (High Risk, Moderate Risk, Low Risk, Prime) based on credit score, mirroring how banks commonly bucket applicants for underwriting.

Suggested libraries used: ggplot2, dplyr, plotly (R)

5.1 Histogram – Loan Amount Distribution

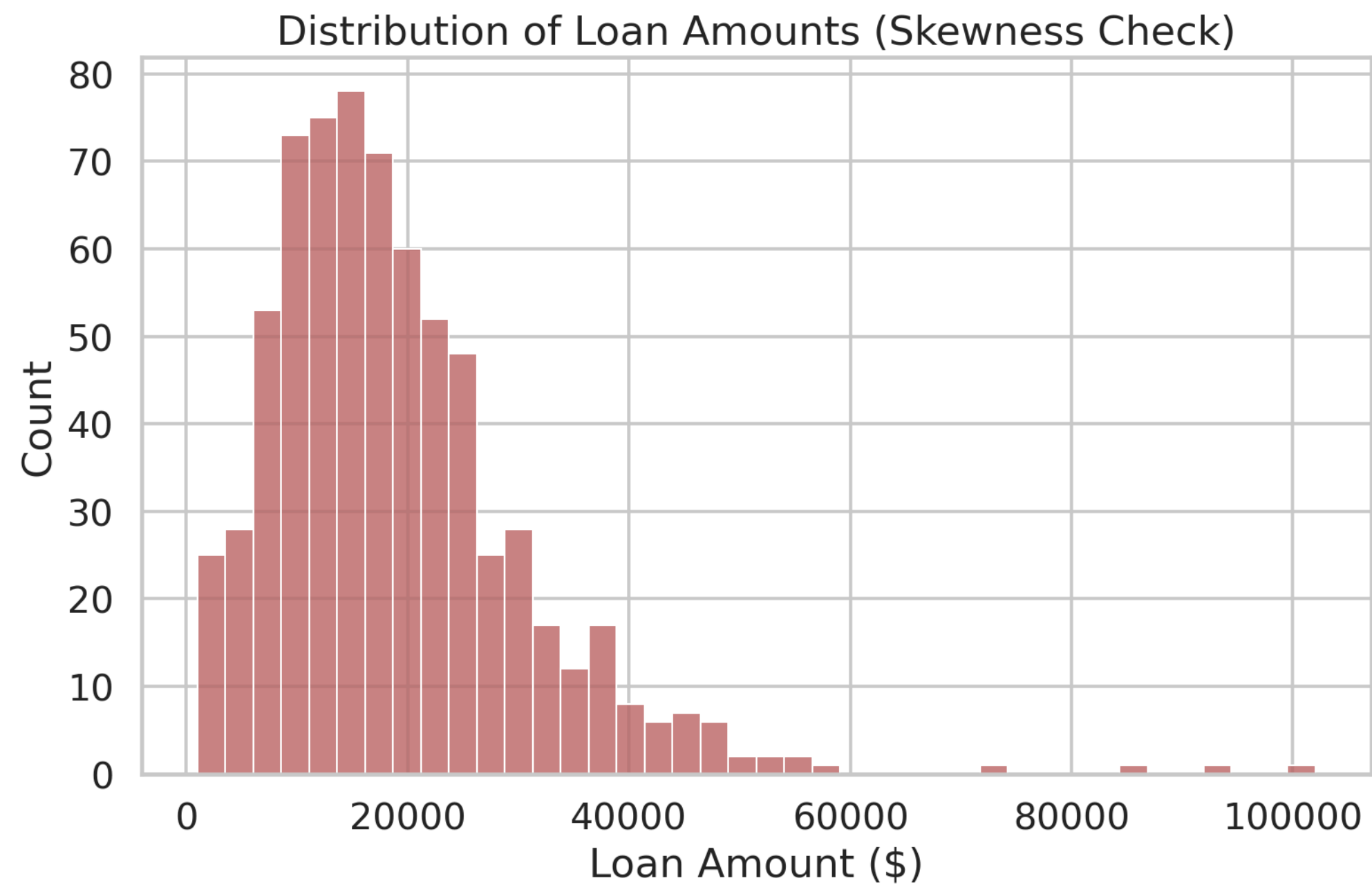


Figure 5.1: Distribution of loan amounts, highlighting right-skewness.

5.2 Box Plot – Loan Amount by Risk Segment

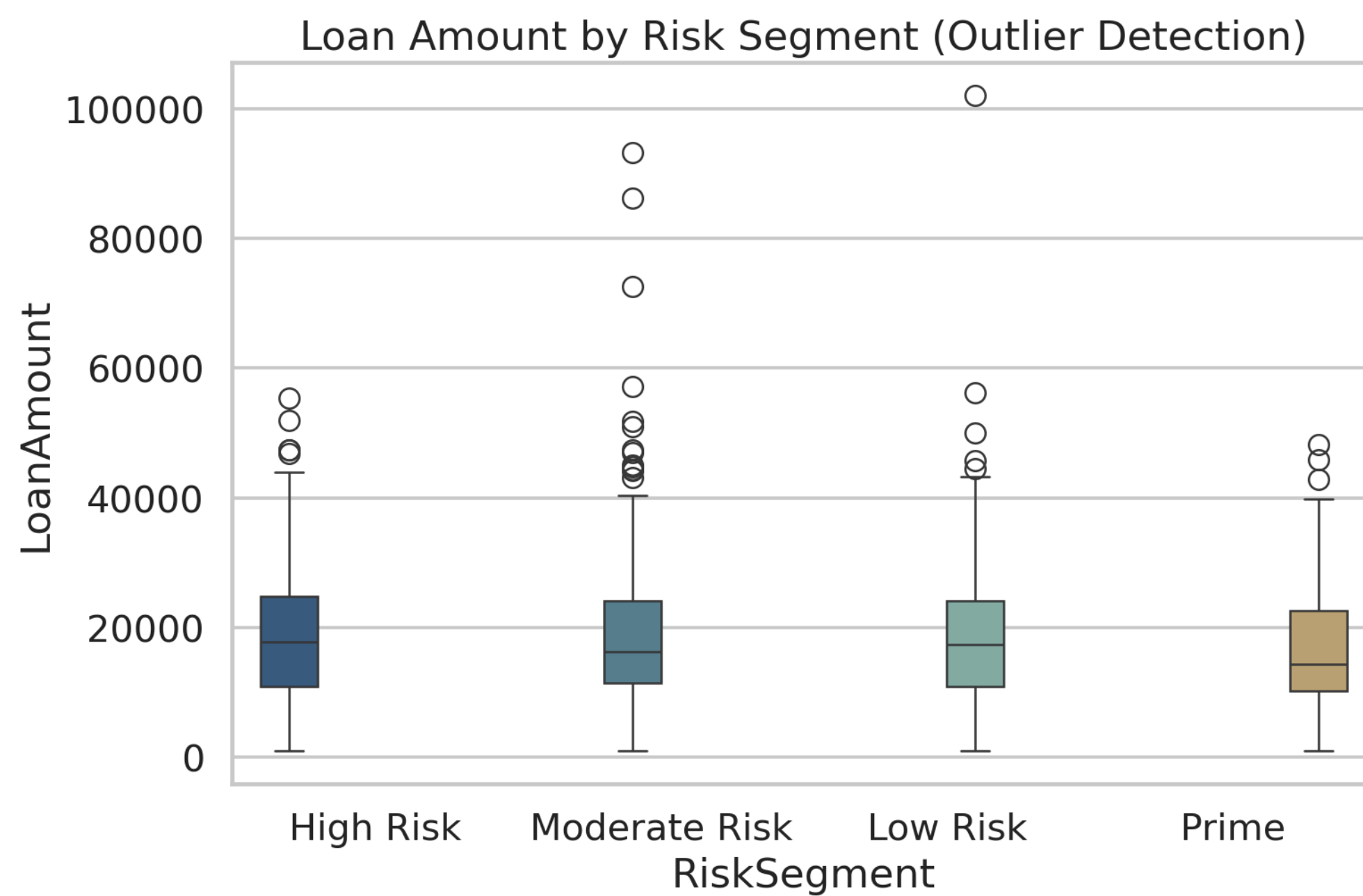


Figure 5.2: Median, spread and outlier loans within each risk segment.

5.3 Density Plot – Credit Score Distribution

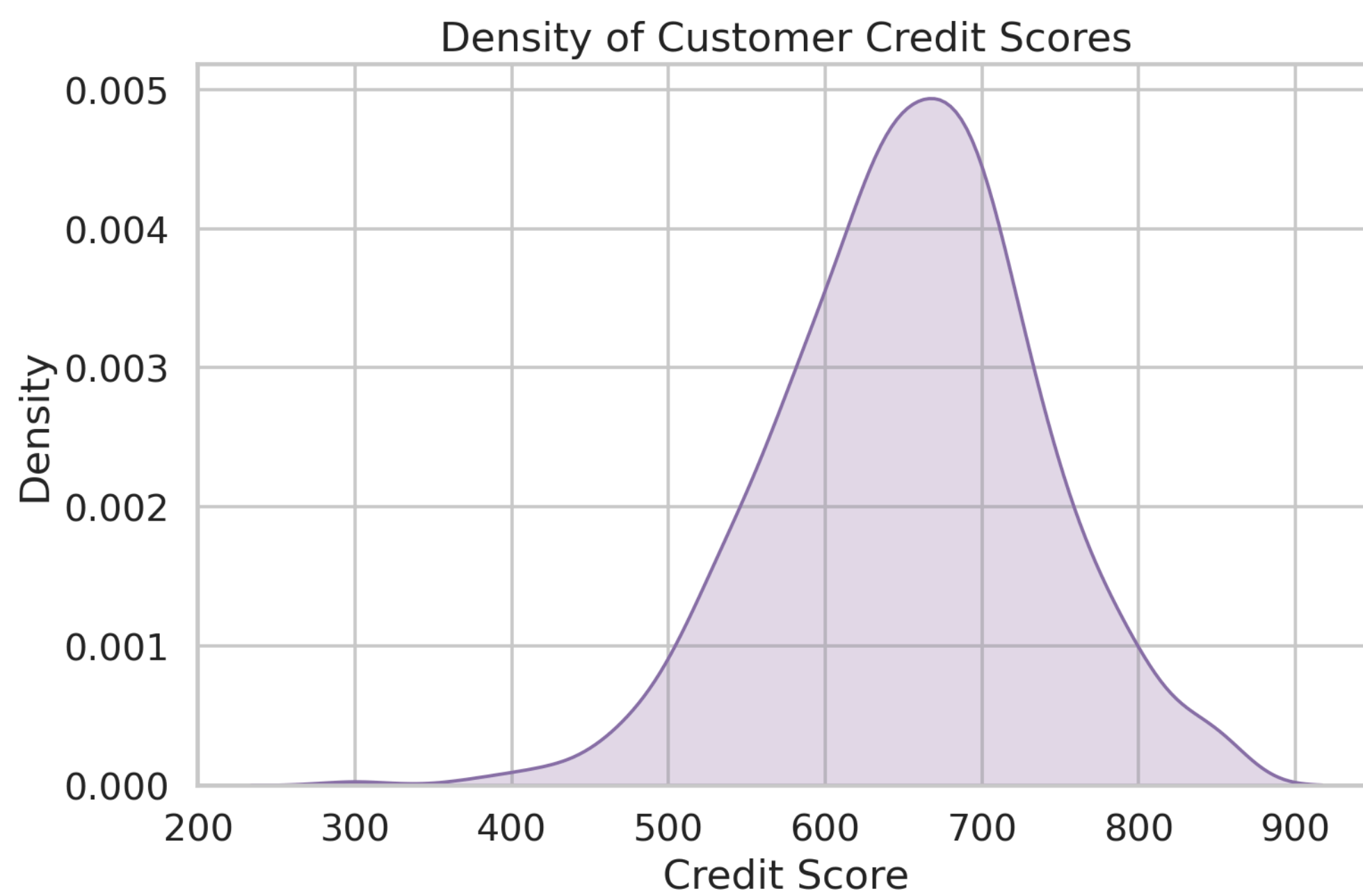


Figure 5.3: Smoothed distribution of customer credit scores.

5.4 Bar Chart – Customers per Risk Segment

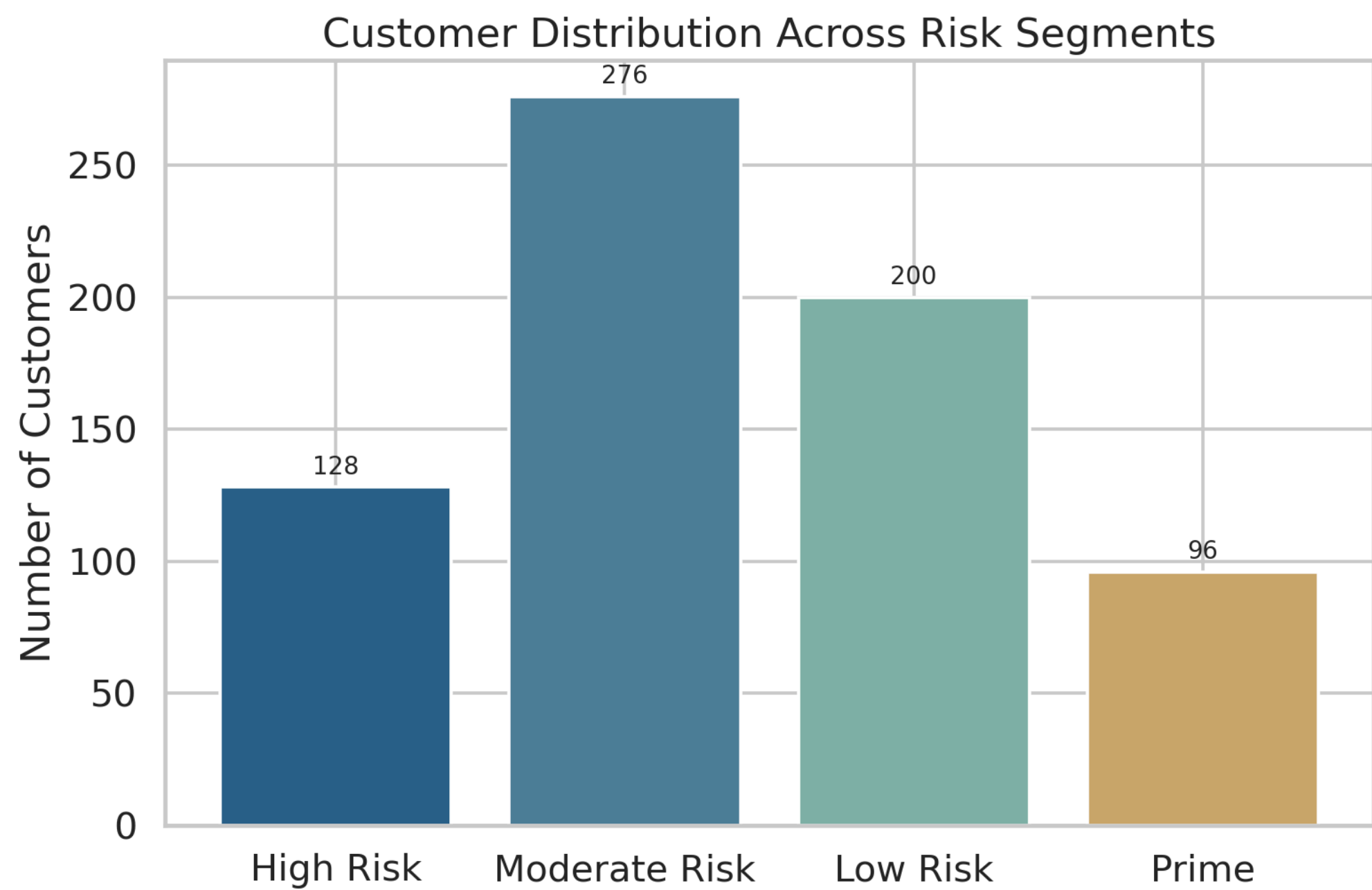


Figure 5.4: Portfolio composition across the four risk segments.

5.5 Supporting Evidence – Income vs Loan Amount

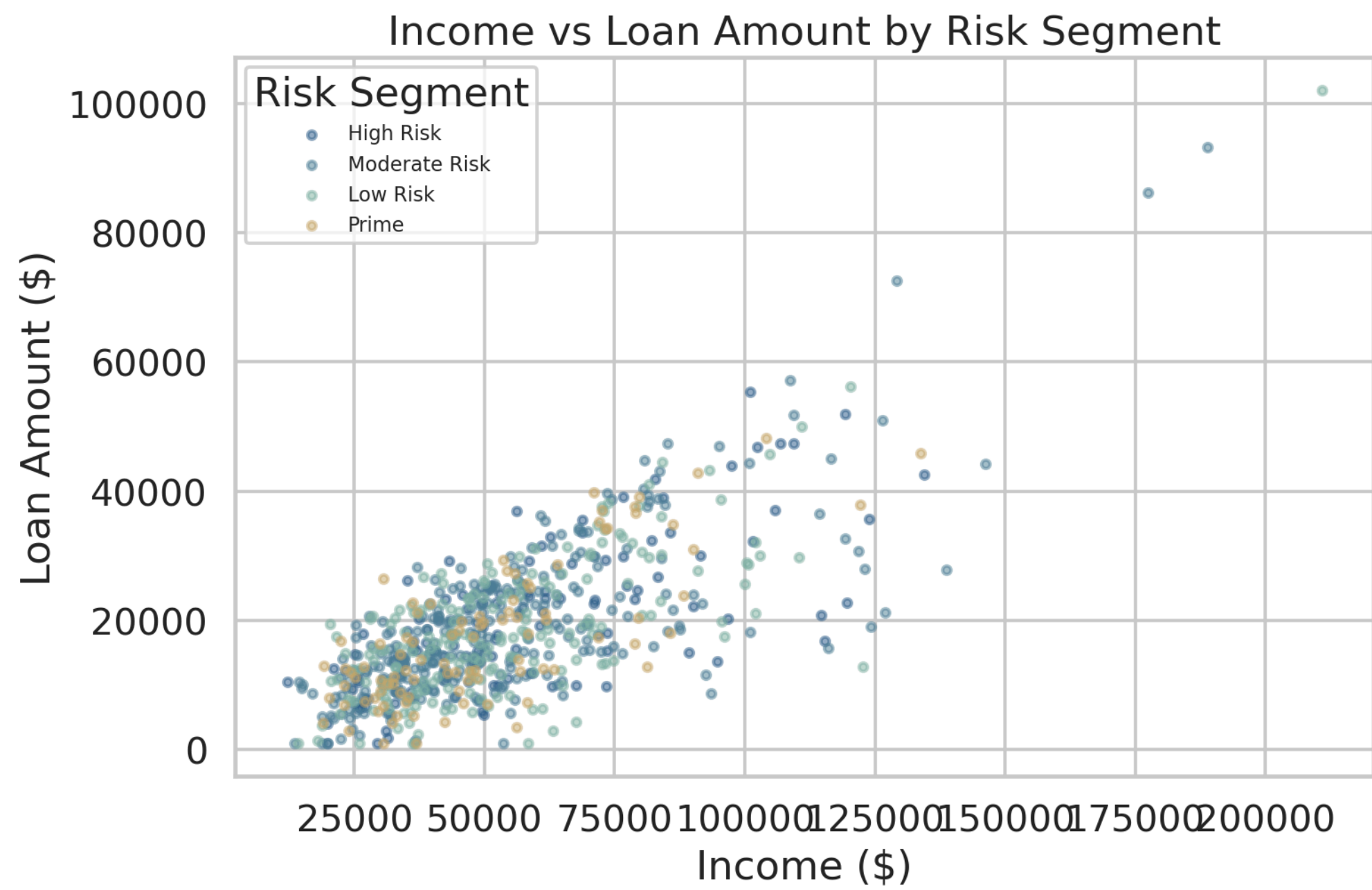


Figure 5.5: Relationship between income and loan amount, coloured by risk segment.

5.6 Evaluation

Which visualization best supports loan approval decisions?

- The box plot by risk segment (Fig. 5.2) is the STRONGEST choice for loan approval decisions: it exposes loan-amount OUTLIERS within each segment – these are the specific applications that most need manual underwriting review, especially within High Risk and Moderate Risk bands.

- The histogram (Fig. 5.1) clearly reveals the overall RIGHT-SKEWNESS of loan amounts, which is important context for choosing appropriate statistical thresholds (e.g. median rather than mean) in a risk model.
- The density plot of credit scores (Fig. 5.3) helps validate that the chosen segment boundaries (580 / 670 / 740) fall in reasonable places relative to the natural spread of scores.
- The bar chart of segment counts (Fig. 5.4) is essential for PORTFOLIO COMPOSITION reporting – e.g. confirming the bank is not over-exposed to High Risk customers.
- The supporting scatter plot (Fig. 5.5) shows loan amount scales with income across all segments, but Prime and Low Risk customers are approved for larger loans at the same income level, evidencing the risk patterns the segments are meant to capture.

Recommendation: Recommend the segmented box plot (Fig. 5.2) as the primary chart for the management dashboard, since it most directly supports the underwriting decision task (spotting risky, outlier-heavy segments); pair it with the bar chart of segment counts (Fig. 5.4) for portfolio-level risk exposure reporting.

6. Overall Conclusion

Across all five scenarios, a consistent pattern emerges: box plots are the most reliable general-purpose choice whenever the underlying task involves comparing variability and identifying outliers across categories (hospital departments, product categories, academic departments, and loan risk segments), because they compactly encode the median, interquartile range and outliers in one view. Bar charts remain the best choice whenever the audience is non-technical (executives, leadership dashboards) or when totals/rankings/counts are the primary message. Density and violin plots add distribution-shape detail that is valuable for analysts but less immediately readable for general audiences. The one scenario where a chart type other than the box plot was recommended as primary was the climate study (Scenario 4), because preserving the calendar/time axis was more important than a compact variability summary — a reminder that the 'best' chart always depends on what specific question the audience needs answered, not on a fixed hierarchy of chart types.

All charts in this report were produced in R using ggplot2 as the core plotting grammar, dplyr for data preparation, treemapify for the treemap in Scenario 1, and plotly to generate interactive dashboard-ready versions of the key recommended charts. The full, self-contained source code is provided in the accompanying file Assignment.R.